Computer Networking

Complete Notes: from zero to CCNA level and beyond

Fundamentals · OSI & TCP/IP · Cabling · Ethernet · IPv4 & IPv6 Subnetting
VLANs · STP · EtherChannel · Routing · OSPF · FHRP
IP Services · Wireless · Security · WAN · SDN · Automation · Cloud

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1. Network Roles & Career Paths

1.1 Who works on networks

Role	What they do every day	Key skills
Network Support / NOC Engineer	Monitors the network 24/7, handles alerts and tickets, first-level troubleshooting	Monitoring tools, ping/traceroute, basic switching and routing
Network Engineer	Designs, configures and maintains switches, routers, firewalls, Wi-Fi and WAN links	VLANs, STP, OSPF/BGP, NAT, ACLs, VPN, Cisco IOS
Network Security Engineer	Firewalls, VPNs, segmentation, intrusion prevention, access control	ACLs, NGFW, IPsec, 802.1X, zero trust
Wireless Engineer	Designs and tunes Wi-Fi coverage and security	802.11, RF, WLC, WPA3, site surveys
Network Automation Engineer	Automates network changes and checks with code	Python, Ansible, APIs, NETCONF/RESTCONF, Git
Cloud Network Engineer	Builds networks in AWS/Azure/GCP and hybrid links to offices	VPC, subnets, routing, VPN, Direct Connect, Transit Gateway
DevOps / SRE / Platform Engineer	Uses networking to run apps: DNS, load balancers, TLS, firewalls, Kubernetes networking	TCP/IP, DNS, HTTP, troubleshooting, cloud networking

1.2 Certifications (in typical order)

Certification	Level	Covers
CompTIA Network+	Entry, vendor-neutral	Networking basics
Cisco CCNA (200-301)	Entry/associate, industry standard	Everything in these notes
AWS Advanced Networking Specialty	Cloud networking	VPC design, hybrid connectivity
Cisco CCNP Enterprise	Professional	Advanced routing, switching, wireless, automation
Cisco DevNet Associate	Automation	Network APIs, Python, automation

CCNA exam domains: Network Fundamentals · Network Access · IP Connectivity · IP Services · Security Fundamentals · Automation and Programmability.

1.3 Practice labs (free)

Tool	Use
Cisco Packet Tracer	Simulated Cisco routers/switches. Best for CCNA beginners (free with a Cisco Networking Academy account)
GNS3 / EVE-NG	Emulate real network operating systems
Containerlab	Network labs with containers (modern, automation-friendly)

Tool	Use
Wireshark	See real packets

2. Network Fundamentals

2.1 What a network is

A **network** is two or more devices connected to share data and resources. Devices that send and receive data are **hosts/endpoints** (PCs, phones, servers, printers, IoT).

2.2 Network types

Type	Meaning	Example
LAN	Local Area Network: one building or campus	Office floor
WLAN	Wireless LAN	Office Wi-Fi
WAN	Wide Area Network: connects LANs over long distances	Head office ↔ branches
MAN	Metropolitan Area Network: across a city	City-wide fibre
SAN	Storage Area Network: servers ↔ storage	Data center storage
Internet	Global network of networks	—
Intranet / Extranet	Private company network / partly opened to partners	Employee portal

2.3 Network devices

Device	Layer	Job
Hub (obsolete)	1	Repeats bits to every port. One collision domain
Switch (L2)	2	Forwards frames by MAC address inside a LAN
Multilayer switch (L3)	2 + 3	Switching plus routing between VLANs
Router	3	Forwards packets between networks by IP address
Firewall / NGFW	3–7	Allows/blocks traffic by rules; NGFW also inspects applications and users
IPS / IDS	3–7	Detects (IDS) or blocks (IPS) attacks
Access Point (AP)	1–2	Connects wireless devices to the wired LAN
Wireless LAN Controller (WLC)	—	Centrally manages many APs
Load balancer	4–7	Spreads traffic across servers
Modem / ONT	1	Connects to the ISP line (DSL, cable, fibre)
PoE switch	—	Powers phones, APs and cameras over the Ethernet cable

2.4 Topologies

Topology	Description	Note
Star	All devices connect to a central switch	Most common LAN design

Topology	Description	Note
Full mesh	Every device connects to every other	Highly redundant, expensive
Partial mesh	Some redundant links	Common in WANs
Bus / Ring	Shared cable / loop	Legacy
Hybrid	Mix of the above	Real networks
Point-to-point	One link between two devices	WAN links

2.5 Network architectures

```
THREE-TIER CAMPUS                    SPINE-LEAF (DATA CENTER)
      [Core]  [Core]                   [Spine1]    [Spine2]
        |   \/   |                      |  \  /  \  /  |
   [Distribution] [Distribution]       [Leaf1] [Leaf2] [Leaf3]
     /     \        /     \              |       |       |
 [Access] [Access] [Access] [Access]   servers  servers  servers
    |        |
  users    users
```

Architecture	Where	Idea
Two-tier (collapsed core)	Small/medium campus	Core and distribution combined
Three-tier	Large campus	Access (users) → Distribution (policies, routing between VLANs) → Core (fast backbone)
Spine-leaf	Data centers	Every leaf connects to every spine: same number of hops between any two servers, easy to scale
SOHO	Small office/home	One device = router + switch + Wi-Fi + firewall
WAN	Companies with sites	Hub-and-spoke, full/partial mesh, SD-WAN
On-premises vs cloud	—	Your own data center vs provider networks (VPC/VNet)

2.6 Traffic types

Type	Sent to	Example
Unicast	One device	Web browsing
Broadcast	All devices in the local network (`FF:FF:FF:FF:FF:FF` , `255.255.255.255`)	ARP request, DHCP Discover
Multicast	A group of interested devices (`224.0.0.0/4`)	OSPF hellos, video streaming
Anycast	The nearest of several devices with the same address	DNS root servers, CDNs

2.7 Bandwidth, throughput, latency

Term	Meaning
Bandwidth	Maximum capacity of a link (e.g. 1 Gbps)
Throughput	Actual data rate achieved
Latency	Time for data to travel (ms). Round-trip time (RTT) = there and back
Jitter	Variation in latency (bad for voice/video)
Packet loss	% of packets that never arrive

Units: **b = bit, B = byte (8 bits)**. Link speeds are in bits (Mbps); file sizes usually in bytes (MB).

3. OSI & TCP/IP Models

3.1 The layers

#	OSI layer	Data unit (PDU)	Address / key info	Devices / protocols
7	Application	Data	—	HTTP, DNS, SSH, SMTP, FTP, SNMP
6	Presentation	Data	Encoding, encryption	TLS, JPEG, ASCII
5	Session	Data	Sessions	RPC, sockets
4	Transport	**Segment** (TCP) / Datagram (UDP)	**Port numbers**	TCP, UDP
3	Network	**Packet**	**IP address**	IP, ICMP, OSPF, routers
2	Data link	**Frame**	**MAC address**	Ethernet, 802.11, ARP, switches
1	Physical	**Bits**	Signals	Cables, fibre, radio, hubs

Memory trick (7 → 1): **A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing.

TCP/IP model: Application (OSI 5–7) · Transport (4) · Internet (3) · Network Access / Link (1–2).

3.2 Encapsulation

```
Sender (going down)                      Receiver (going up)
Data                                     Data
[TCP hdr | Data]           = Segment     ↑ remove TCP header
[IP hdr | TCP hdr | Data]  = Packet      ↑ remove IP header
[Eth hdr | IP | TCP | Data | FCS] = Frame ↑ check FCS, remove Ethernet header
010110101...               = Bits        ↑
```

- **Encapsulation:** each layer adds its header on the way down. **De-encapsulation:** removed on the way up.
- **Key rule:** while a packet crosses networks, the **IP addresses stay the same** end to end (unless NAT), but the **MAC addresses change at every router hop**.

3.3 Important headers (simplified)

Header	Key fields
Ethernet	Destination MAC, Source MAC, Type (0x0800 = IPv4, 0x86DD = IPv6, 0x8100 = VLAN tag), FCS (error check)
IPv4	Version, TTL (decreases by 1 at each router; at 0 the packet is dropped), Protocol (6 = TCP, 17 = UDP, 1 = ICMP), Source IP, Destination IP
TCP	Source/destination port, sequence number, acknowledgment number, flags (SYN, ACK, FIN, RST), window size
UDP	Source/destination port, length, checksum (only 8 bytes)

4. Physical Layer: Cabling & Interfaces

4.1 Copper and fibre

Media	Distance	Notes
UTP Cat5e	100 m	1 Gbps
UTP Cat6 / Cat6a	100 m	10 Gbps (Cat6 up to ~55 m, Cat6a full 100 m)
Multimode fibre (MMF)	Hundreds of metres	Cheaper, LED/VCSEL light, inside buildings/data centers
Single-mode fibre (SMF)	Kilometres	Laser, long distance, more expensive
Wireless	Tens of metres	Shared medium, half duplex

Connectors: RJ45 (copper), LC/SC (fibre). Transceivers: SFP (1G), SFP+ (10G), QSFP (40/100G). Straight-through vs crossover cables are mostly irrelevant today because of **Auto-MDIX**.

4.2 Ethernet standards

Standard	Speed	Media
100BASE-TX	100 Mbps (Fast Ethernet)	Copper
1000BASE-T	1 Gbps	Copper
1000BASE-LX/SX	1 Gbps	Fibre (LX single/multi-mode, SX multimode)
10GBASE-T / -SR / -LR	10 Gbps	Copper / short fibre / long fibre

4.3 Speed and duplex

- **Half duplex:** send or receive, not both (old hubs, Wi-Fi). Uses CSMA/CD on wired.
- **Full duplex:** send and receive at once. No collisions.
- **Duplex mismatch** (one side full, other half) → late collisions, slow, errors. Use auto-negotiation on both sides or set both the same.

4.4 Interface errors (`show interfaces`)

Counter	Meaning / likely cause
CRC / input errors	Corrupted frames: bad cable, interference, bad port
Runts / giants	Frames too small (<64 bytes) / too big (>1518 bytes): collisions, MTU problems
Collisions / late collisions	Half duplex or duplex mismatch
Output drops	Congested interface
Interface status `up/down`	Physical OK but line protocol not (encapsulation, keepalives)
`administratively down`	Interface is `shutdown` — run `no shutdown`
`err-disabled`	Shut down by a protection feature (port security, BPDU Guard)

5. Ethernet, MAC Addresses & Switching

5.1 MAC addresses

- 48 bits, written in hex: `00:1A:2B:3C:4D:5E` or Cisco style `001a.2b3c.4d5e`.
- First half = **OUI** (vendor ID), second half = device-specific.
- Broadcast MAC: `FFFF.FFFF.FFFF`.

5.2 How a switch forwards frames

```
1. LEARN    : frame arrives on port 1 from MAC A → switch records "A is on port 1"
2. FORWARD  : destination MAC known → send only out that port
3. FLOOD    : destination unknown or broadcast → send out all ports except the one it came from
4. FILTER   : destination is on the same port it came from → drop
```

- The **MAC address table** (CAM table) entries age out after 300 seconds by default.
- Commands: `show mac address-table`, `clear mac address-table dynamic`.

5.3 Collision vs broadcast domains

Device	Collision domains	Broadcast domains
Hub	One for all ports	One
Switch	One **per port**	One (per VLAN)
Router	One per port	One **per interface** (routers stop broadcasts)

5.4 ARP (Address Resolution Protocol)

Finds the MAC address for a known IPv4 address on the local network.

```
PC1 (10.0.0.10) wants to reach 10.0.0.20:
  ARP Request  (broadcast): "Who has 10.0.0.20? Tell 10.0.0.10"
  ARP Reply    (unicast):   "10.0.0.20 is at aaaa.bbbb.cccc"
  PC1 stores it in its ARP cache.
If the destination is in ANOTHER network, PC1 ARPs for its DEFAULT GATEWAY's MAC instead.
```

Commands: `arp -a` (Windows), `ip neigh` (Linux), `show ip arp` (Cisco).

6. Binary, IPv4 Addressing & Subnetting

6.1 Number systems

System	Base	Digits	Used for
Decimal	10	0–9	How people write IPv4: `192.168.1.10`
Binary	2	0, 1	How computers actually store IPs and masks
Hexadecimal (hex)	16	0–9, A–F	MAC addresses, IPv6

Unit	Size
Bit	One 0 or 1
Nibble	4 bits (= one hex digit)
Byte / **octet**	8 bits. An IPv4 address = 4 octets = 32 bits

6.2 Place values (the key to everything)

Each position in an 8-bit octet has a value. **Memorize this row:**

Position	8th	7th	6th	5th	4th	3rd	2nd	1st
Power of 2	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Value	**128**	**64**	**32**	**16**	**8**	**4**	**2**	**1**

All 8 bits on: 128+64+32+16+8+4+2+1 = **255** → that's why an octet goes from 0 to 255.

Powers of 2 (for counting addresses):

2^1	2^2	2^3	2^4	2^5	2^6	2^7	2^8	2^9	2^{10}	2^{11}	2^{12}	2^{16}
2	4	8	16	32	64	128	256	512	1024	2048	4096	65536

6.3 Binary → decimal

Write the place values above the bits and **add the values where the bit is 1**.

Binary	128	64	32	16	8	4	2	1	Sum
11000000	1	1	0	0	0	0	0	0	128+64 = **192**
10101000	1	0	1	0	1	0	0	0	128+32+8 = **168**
00001010	0	0	0	0	1	0	1	0	8+2 = **10**
01001101	0	1	0	0	1	1	0	1	64+8+4+1 = **77**

6.4 Decimal → binary (subtraction method)

Go left to right. **If the number is ≥ the place value: write 1 and subtract. Otherwise write 0.**

Example: 172

Place value	128	64	32	16	8	4	2	1
Fits?	172 ≥ 128 ✓	44 < 64 ✗	44 ≥ 32 ✓	12 < 16 ✗	12 ≥ 8 ✓	4 ≥ 4 ✓	0 ✗	0 ✗
Remaining	44	44	12	12	4	0	0	0
Bit	**1**	**0**	**1**	**0**	**1**	**1**	**0**	**0**

172 = **10101100**. Check: 128+32+8+4 = 172 ✓

Example: 200 → 128 ✓ (72 left) → 64 ✓ (8 left) → 32 ✗ → 16 ✗ → 8 ✓ (0 left) → 4 ✗ → 2 ✗ → 1 ✗ = **11001000**

Full address:

```
192      .168      .10       .77
11000000 .10101000 .00001010 .01001101
```

6.5 Hexadecimal

One hex digit = 4 bits. A=10, B=11, C=12, D=13, E=14, F=15.

Hex	Binary	Hex	Binary	Hex	Binary	Hex	Binary
0	0000	4	0100	8	1000	C	1100
1	0001	5	0101	9	1001	D	1101
2	0010	6	0110	A	1010	E	1110
3	0011	7	0111	B	1011	F	1111

- **Hex → decimal:** `1A` = 1×16 + 10 = **26**. `FF` = 15×16 + 15 = **255**. `2F` = 2×16 + 15 = **47**.
- **Hex → binary:** convert each digit separately: `0DB8` = 0000 1101 1011 1000.
- Used in MAC addresses (`00:1A:2B:...`) and IPv6 (`2001:db8::1`).

6.6 The subnet mask in binary

An IPv4 address has a **network part** (which network) and a **host part** (which device). The **subnet mask** marks them: **1 = network bit, 0 = host bit**. The 1s are always continuous from the left.

```
/24  = 11111111.11111111.11111111.00000000 = 255.255.255.0     (24 network bits, 8 host bits)
/27  = 11111111.11111111.11111111.11100000 = 255.255.255.224   (27 network bits, 5 host bits)
```

Prefix length (`/27`) = number of 1s. **Host bits (h)** = 32 − prefix.

Only these values can appear in a mask octet:

Mask octet	Binary	Network bits in octet	Block size (256 − value)
0	00000000	0	256
128	10000000	1	128
192	11000000	2	64
224	11100000	3	32

Mask octet	Binary	Network bits in octet	Block size (256 − value)
240	11110000	4	16
248	11111000	5	8
252	11111100	6	4
254	11111110	7	2
255	11111111	8	1

The formulas:

Want to know	Formula	Example /27 (h = 5)
Total addresses in a subnet	2^h	$2^5 = 32$
Usable hosts	$2^h - 2$ (network + broadcast reserved)	$32 - 2 = 30$
Number of subnets when borrowing s bits	2^s	/24 → /27 borrows 3 bits → $2^3 = 8$ subnets

6.7 How the network address is found: the AND operation

Routers find the network by doing a bitwise **AND** of the IP and the mask. **AND rule: 1 AND 1 = 1, anything else = 0.**

Example: 192.168.10.77/27 (only the last octet changes, because the first three mask octets are 255)

```
IP   last octet   77 = 0 1 0 | 0 1 1 0 1
Mask last octet  224 = 1 1 1 | 0 0 0 0 0
                       network | host bits
AND result           = 0 1 0 | 0 0 0 0 0 = 64   → Network   192.168.10.64

Set all host bits to 1 = 0 1 0 | 1 1 1 1 1 = 95   → Broadcast 192.168.10.95
```

Result	Value
Network address (all host bits 0)	192.168.10.64
First usable host (network + 1)	192.168.10.65
Last usable host (broadcast − 1)	192.168.10.94
Broadcast address (all host bits 1)	192.168.10.95
Usable hosts	$2^5 - 2 = 30$

6.8 The fast method: block size

Use this in exams and real work. No binary needed once you understand 6.7.

1. **Find the interesting octet:** the octet where the mask is **not** 255 or 0.

Prefix	/1–/8	/9–/16	/17–/24	/25–/32
Interesting octet	1st	2nd	3rd	4th

2. **Block size** = 256 − mask value in that octet.
3. Subnets start at **multiples of the block size**: 0, block, 2×block...
4. **Network** = the largest multiple that is ≤ the IP's value in that octet. Octets to the right become 0.
5. **Broadcast** = next network − 1 (octets to the right become 255).

Example 1: 10.20.30.200/26 - /26 → 4th octet, mask 192 → block **64** → subnets .0, .64, .128, **.192** - 200 is in the .192 block → **Network 10.20.30.192**, broadcast **10.20.30.255**, hosts .193 – .254 (62 hosts)

Example 2: 172.16.45.10/20 - /20 → 3rd octet, mask 255.255.**240**.0 → block **16** → 0, 16, **32**, 48... - 45 is in the 32 block → **Network 172.16.32.0**, broadcast **172.16.47.255** - Hosts 172.16.32.1 – 172.16.47.254 → $2^{12} - 2$ = **4,094**

Example 3: 192.168.5.130/30 - block 4 → .128, .132 → **Network .128**, hosts **.129, .130**, broadcast **.131** (typical router-to-router link)

Example 4: 10.5.200.9/14 - /14 → 2nd octet, mask 255.**252**.0.0 → block **4** → 0, **4**, 8... - 5 is in the 4 block → **Network 10.4.0.0**, broadcast **10.7.255.255**, $2^{18} - 2$ = 262,142 hosts

6.9 Designing subnets

A) "I need N subnets" → find the smallest s where $2^s \geq N$, add s to the prefix.

Need 6 subnets from 192.168.1.0/24 → $2^3 = 8 \geq 6$ → borrow 3 bits → **/27** → block 32, 30 hosts each:

#	Network	Usable range	Broadcast
1	192.168.1.0/27	.1 – .30	.31
2	192.168.1.32/27	.33 – .62	.63
3	192.168.1.64/27	.65 – .94	.95
4	192.168.1.96/27	.97 – .126	.127
5	192.168.1.128/27	.129 – .158	.159
6	192.168.1.160/27	.161 – .190	.191
7–8	.192/27, .224/27	spare	

B) "I need H hosts per subnet" → find the smallest h where $2^h - 2 \geq H$, prefix = 32 − h.

Hosts needed (up to)	2	6	14	30	62	126	254	510	1022	2046	4094
Prefix	/30	/29	/28	/27	/26	/25	/24	/23	/22	/21	/20

Need 500 hosts → $2^9 - 2 = 510 \geq 500$ → h = 9 → **/23**.

6.10 Address classes (historic) and special ranges

Class	First octet	Default mask	Note
A	1–126	/8	127 = loopback
B	128–191	/16	
C	192–223	/24	
D	224–239	—	Multicast

Class	First octet	Default mask	Note
E	240–255	—	Experimental

Classes are no longer used for routing (we use CIDR), but the terms still appear.

Range	Purpose
10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16	Private (RFC 1918)
127.0.0.0/8	Loopback
169.254.0.0/16	APIPA / link-local (no DHCP answer)
100.64.0.0/10	Carrier-grade NAT (ISP use)
0.0.0.0/0	Default route ("everything")
255.255.255.255	Limited broadcast

6.11 Prefix reference table

Prefix	Mask	Addresses	Usable hosts
/8	255.0.0.0	16,777,216	16,777,214
/16	255.255.0.0	65,536	65,534
/20	255.255.240.0	4,096	4,094
/22	255.255.252.0	1,024	1,022
/23	255.255.254.0	512	510
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2 (point-to-point links)
/31	255.255.255.254	2	2 (point-to-point, RFC 3021)
/32	255.255.255.255	1	Single host / loopback

6.12 VLSM (Variable Length Subnet Masking)

Give each subnet only the size it needs. **Allocate the largest first**, so the blocks line up.

Example from `192.168.1.0/24`:

Need	Hosts	Prefix	Subnet	Range	Broadcast
Sales	100	/25	192.168.1.0	.1 – .126	.127
IT	50	/26	192.168.1.128	.129 – .190	.191
Mgmt	20	/27	192.168.1.192	.193 – .222	.223

Need	Hosts	Prefix	Subnet	Range	Broadcast
Link R1–R2	2	/30	192.168.1.224	.225 – .226	.227

6.13 Route summarization (in binary)

Combine contiguous networks into one route. Write the changing octet in binary and **count the bits that are the same in all of them**.

```
10.1.0.0  → 3rd octet 0 = 000000|00
10.1.1.0  → 3rd octet 1 = 000000|01
10.1.2.0  → 3rd octet 2 = 000000|10
10.1.3.0  → 3rd octet 3 = 000000|11
Common bits: 8 + 8 + 6 = 22  →  summary 10.1.0.0/22
```

Smaller routing tables and more stability.

6.14 Practice questions (answers below)

1. Convert 150 to binary.
2. Convert 11110000 to decimal.
3. Convert hex 2F to decimal.
4. Network, broadcast and host range of 192.168.50.100/28?
5. How many usable hosts in a /22?
6. Smallest subnet for 50 hosts?
7. Are 10.0.0.70/26 and 10.0.0.130/26 in the same subnet?
8. How many /24 subnets fit in 172.16.0.0/16?
9. Summarize 192.168.16.0/24 to 192.168.23.0/24 into one route.
10. Last usable host of 10.10.10.0/23?

Answers:

#	Answer	How
1	10010110	128 + 16 + 4 + 2 = 150
2	240	128 + 64 + 32 + 16
3	47	2×16 + 15
4	Network .96, hosts .97 – .110, broadcast .111	/28 → block 16 → 96 ≤ 100 < 112
5	1,022	h = 10 → $2^{10} - 2$
6	/26 (62 hosts)	/27 gives only 30
7	No	/26 blocks: 64–127 and 128–191
8	256	Borrow 8 bits → 2^8
9	192.168.16.0/21	16 = 00010000, 23 = 00010111 → 5 common bits → 16 + 5 = 21
10	10.10.11.254	/23 → block 2 in 3rd octet → network 10.10.10.0, broadcast 10.10.11.255

7. IPv6

7.1 Format

- 128 bits, 8 groups of 4 hex digits: `2001:0db8:0000:0000:0000:ff00:0042:8329`
- **Shorten:** drop leading zeros in each group, and replace **one** run of all-zero groups with `::` → `2001:db8::ff00:42:8329`
- Usual prefix for a LAN: **/64** (64 bits network, 64 bits interface ID). No broadcast in IPv6; multicast is used instead.

7.2 Address types

Type	Prefix	Use
Global unicast (GUA)	`2000::/3`	Public, routable on the internet
Unique local (ULA)	`fc00::/7` (in practice `fd00::/8`)	Private, like RFC 1918
Link-local	`fe80::/10`	Only on the local link; every IPv6 interface has one; used by routing protocols and as next hop
Loopback	`::1/128`	This host
Unspecified	`::/128`	"No address yet"
Multicast	`ff00::/8`	`ff02::1` all nodes, `ff02::2` all routers
Anycast	(from unicast space)	Nearest of several devices

7.3 Address assignment

Method	How
Static	Configured manually
EUI-64	Interface ID built from the MAC: split the MAC, insert `FFFE`, flip the 7th bit
SLAAC	Host builds its own address from the router's advertised prefix (Router Advertisement)
Stateless DHCPv6	SLAAC for the address + DHCPv6 for extra info (DNS)
Stateful DHCPv6	DHCPv6 server assigns the address (like DHCPv4)

NDP (Neighbor Discovery Protocol) replaces ARP, using ICMPv6: Neighbor Solicitation/Advertisement (find MACs), Router Solicitation/Advertisement (find routers and prefixes), Duplicate Address Detection.

```
R1(config)# ipv6 unicast-routing
R1(config)# interface g0/0
R1(config-if)# ipv6 address 2001:db8:1:1::1/64
R1(config-if)# ipv6 address fe80::1 link-local
R1(config)# ipv6 route ::/0 2001:db8:0:12::2          ! IPv6 default route
R1# show ipv6 interface brief
```

8. Cisco IOS CLI Essentials

8.1 Modes

```
Router>                          User EXEC (look only)
Router> enable
Router#                          Privileged EXEC (show, debug, save)
Router# configure terminal
Router(config)#                  Global configuration
Router(config)# interface g0/1
Router(config-if)#               Interface configuration
Router(config-if)# exit / end    Up one level / back to Privileged EXEC
```

8.2 Basic device setup

```
Switch(config)# hostname SW1
SW1(config)# enable secret <strong-password>          ! encrypted privileged password
SW1(config)# service password-encryption              ! hide other passwords in the config
SW1(config)# banner motd #Authorized access only#
SW1(config)# no ip domain-lookup                      ! stop treating typos as hostnames
SW1(config)# line console 0
SW1(config-line)# password <pw>
SW1(config-line)# login
SW1(config-line)# logging synchronous
SW1(config)# interface vlan 1                         ! management IP on a switch (SVI)
SW1(config-if)# ip address 192.168.1.2 255.255.255.0
SW1(config-if)# no shutdown
SW1(config)# ip default-gateway 192.168.1.1
SW1# copy running-config startup-config               ! SAVE (or: write memory)
```

Config	Stored in	Meaning
running-config	RAM	Active now; lost on reboot unless saved
startup-config	NVRAM	Loaded at boot

8.3 SSH instead of Telnet

```
SW1(config)# ip domain-name lab.local
SW1(config)# crypto key generate rsa modulus 2048
SW1(config)# ip ssh version 2
SW1(config)# username admin secret <strong-password>
SW1(config)# line vty 0 15
SW1(config-line)# login local
SW1(config-line)# transport input ssh                 ! no Telnet (plain text)
```

8.4 Most-used show commands

Command	Shows
`show running-config`	Current configuration
`show ip interface brief`	Interfaces, IPs, up/down status (first command to run)

Command	Shows
`show interfaces g0/1`	Detailed counters, errors, speed/duplex
`show interfaces status`	Switch ports: VLAN, speed, duplex, status
`show ip route`	Routing table
`show vlan brief`	VLANs and their ports
`show interfaces trunk`	Trunk ports and allowed VLANs
`show mac address-table`	Learned MACs
`show spanning-tree`	STP root, port roles/states
`show cdp neighbors detail` / `show lldp neighbors`	Directly connected devices
`show ip ospf neighbor`	OSPF neighbors
`show ip nat translations`	NAT entries
`show access-lists`	ACLs and hit counters
`show version`	IOS version, uptime, hardware
`show logging`	Log messages

Tips: `?` for help, `Tab` to complete, `do show ...` from config mode, `| include`, `| section` to filter output.

9. VLANs & Trunking

9.1 VLAN

- **What:** a **V**irtual **LAN** splits one physical switch into several separate logical networks (separate broadcast domains).
- **Why:** security (separate HR, guests, servers), less broadcast traffic, organization by function instead of location.
- **Rule:** one VLAN = one IP subnet. Devices in different VLANs need a **router or L3 switch** to talk.

```
SW1(config)# vlan 10
SW1(config-vlan)# name SALES
SW1(config)# vlan 20
SW1(config-vlan)# name HR
SW1(config)# interface range f0/1 - 10
SW1(config-if-range)# switchport mode access
SW1(config-if-range)# switchport access vlan 10
SW1(config)# interface f0/11
SW1(config-if)# switchport mode access
SW1(config-if)# switchport access vlan 20
SW1(config-if)# switchport voice vlan 100             ! IP phone traffic tagged in VLAN 100
```

9.2 Trunks (802.1Q)

- **Access port:** belongs to one VLAN, frames are untagged (for end devices).
- **Trunk port:** carries many VLANs between switches/routers. Frames get an **802.1Q tag** (4 bytes, includes the 12-bit VLAN ID: 1–4094).
- **Native VLAN:** frames in this VLAN cross the trunk untagged (default VLAN 1). Best practice: change it to an unused VLAN and make it match on both ends.

```
SW1(config)# interface g0/1
SW1(config-if)# switchport trunk encapsulation dot1q    ! only needed on some switches
SW1(config-if)# switchport mode trunk
SW1(config-if)# switchport trunk allowed vlan 10,20,100
SW1(config-if)# switchport trunk native vlan 999
SW1(config-if)# switchport nonegotiate                  ! disable DTP
```

DTP (Dynamic Trunking Protocol) negotiates trunks automatically — disable it for security. **VTP** shares VLAN lists between switches — usually set to `vtp mode transparent` or off to avoid accidental VLAN deletion.

9.3 Inter-VLAN routing

Router-on-a-stick (one router link, subinterfaces):

```
R1(config)# interface g0/0.10
R1(config-subif)# encapsulation dot1Q 10
R1(config-subif)# ip address 192.168.10.1 255.255.255.0
R1(config)# interface g0/0.20
R1(config-subif)# encapsulation dot1Q 20
R1(config-subif)# ip address 192.168.20.1 255.255.255.0
R1(config)# interface g0/0
R1(config-if)# no shutdown
```

Layer 3 switch with SVIs (faster, common today):

```
SW1(config)# ip routing
SW1(config)# interface vlan 10
SW1(config-if)# ip address 192.168.10.1 255.255.255.0
SW1(config-if)# no shutdown
SW1(config)# interface vlan 20
SW1(config-if)# ip address 192.168.20.1 255.255.255.0
SW1(config-if)# no shutdown
```

Hosts in VLAN 10 use `192.168.10.1` as their default gateway.

10. Spanning Tree Protocol (STP)

10.1 Why

Redundant links between switches create **Layer 2 loops** → broadcast storms, MAC table instability, network down in seconds (Ethernet frames have no TTL). **STP** blocks redundant paths and re-opens them if the active path fails.

10.2 How it works

1. **Elect the root bridge:** lowest **Bridge ID** = priority (default 32768 + VLAN ID) then lowest MAC.
2. **Root port** on every non-root switch: the port with the lowest path cost to the root.
3. **Designated port** on every segment: the side with the lowest cost to the root.
4. All other ports → **blocking** (alternate).

Link speed	STP cost (short)
10 Mbps	100
100 Mbps	19
1 Gbps	4
10 Gbps	2

Version	Note
802.1D STP	Original, slow (30–50 s convergence)
PVST+	Cisco: one STP per VLAN
RSTP (802.1w) / Rapid PVST+	Converges in about a second. Port roles: root, designated, alternate, backup. States: discarding, learning, forwarding
MSTP (802.1s)	Groups VLANs into a few instances

```
SW1(config)# spanning-tree mode rapid-pvst
SW1(config)# spanning-tree vlan 10,20 root primary      ! make this switch the root
SW1(config)# interface range f0/1 - 24
SW1(config-if-range)# spanning-tree portfast             ! end-device ports: forward immediately
SW1(config-if-range)# spanning-tree bpduguard enable     ! err-disable if a switch is plugged in
SW1# show spanning-tree vlan 10
```

Best practice: **choose** the root bridge (core/distribution switch), use PortFast + BPDU Guard on access ports.

11. EtherChannel (Link Aggregation)

Bundles several physical links into **one logical link**: more bandwidth and redundancy, and STP sees one link (no blocking).

Protocol	Modes	Note
LACP (802.3ad, open standard)	`active` / `passive`	Recommended
PAgP (Cisco)	`desirable` / `auto`	Cisco only
Static	`on`	No negotiation; risky if misconfigured

```
SW1(config)# interface range g0/1 - 2
SW1(config-if-range)# channel-group 1 mode active        ! LACP
SW1(config)# interface port-channel 1
SW1(config-if)# switchport mode trunk
SW1# show etherchannel summary
```

All member ports must match: speed, duplex, VLAN/trunk settings.

12. Routing

12.1 How a router decides

1. Receives a packet → checks the destination IP.
2. Looks in the **routing table** and picks the **longest prefix match** (most specific route).
3. Forwards out the interface/next hop. Decreases TTL. Rewrites the Layer 2 header (new MACs).
4. No match and no default route → **drop** (and send ICMP "unreachable").

Example: routes `10.0.0.0/8`, `10.1.0.0/16`, `10.1.1.0/24` → a packet to `10.1.1.5` uses `/24`.

12.2 Routing table codes

```
R1# show ip route
C    10.1.1.0/24 is directly connected, GigabitEthernet0/0
L    10.1.1.1/32 is directly connected, GigabitEthernet0/0
S    192.168.50.0/24 [1/0] via 10.1.12.2
O    172.16.0.0/16 [110/3] via 10.1.12.2, 00:05:12, GigabitEthernet0/1
S*   0.0.0.0/0 [1/0] via 203.0.113.1                    ← gateway of last resort
          [AD/metric]
```

C = connected, L = local, S = static, O = OSPF, D = EIGRP, B = BGP, * = default.

12.3 Administrative distance (AD)

When two **different sources** know the same route, the **lower AD wins** (trustworthiness). Within one protocol, the lower **metric** wins.

Source	AD
Connected	0
Static	1
eBGP	20
EIGRP (internal)	90
OSPF	110
RIP	120
iBGP	200
Unusable	255

12.4 Static routes

```
R1(config)# ip route 192.168.50.0 255.255.255.0 10.1.12.2          ! via next-hop IP
R1(config)# ip route 192.168.50.0 255.255.255.0 g0/1 10.1.12.2     ! fully specified
R1(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.1                    ! default route
R1(config)# ip route 0.0.0.0 0.0.0.0 198.51.100.1 5                 ! floating static (backup, AD 5)
```

A **floating static route** has a higher AD, so it is used only if the main route disappears.

12.5 Dynamic routing protocols

Type	How	Examples
Distance vector	Learns routes from neighbors ("routing by rumour")	RIP
Advanced distance vector	Faster, loop-free, Cisco	EIGRP
Link state	Every router builds a full map and runs SPF (Dijkstra)	OSPF, IS-IS
Path vector	Between organizations on the internet	BGP
IGP vs EGP	Inside one organization vs between organizations	OSPF/EIGRP vs BGP

12.6 OSPF (single area, CCNA focus)

Term	Meaning
Router ID	32-bit ID: manual `router-id` → highest loopback IP → highest active interface IP
Area	Group of routers; area 0 = backbone. Single-area OSPF uses area 0
Neighbor / adjacency	Routers discovered by Hello packets (multicast 224.0.0.5)
LSA / LSDB	Link-state advertisements / the database (the "map")
Cost	Metric = reference bandwidth (100 Mbps default) / interface bandwidth. Lower is better
DR / BDR	On Ethernet (broadcast) segments, a Designated Router reduces updates. Highest priority (default 1), then highest router ID. Priority 0 = never DR
Timers	Hello 10 s, Dead 40 s on Ethernet — must match between neighbors

Neighbor states: Down → Init → 2-Way → ExStart → Exchange → Loading → **Full**.

```
R1(config)# router ospf 1
R1(config-router)# router-id 1.1.1.1
R1(config-router)# auto-cost reference-bandwidth 10000    ! makes 1G/10G costs meaningful
R1(config-router)# network 10.1.1.0 0.0.0.255 area 0     ! wildcard mask = inverse of subnet mask
R1(config-router)# network 10.1.12.0 0.0.0.3 area 0
R1(config-router)# passive-interface g0/0                ! no hellos toward user LAN
R1(config-router)# default-information originate         ! advertise this router's default route
! or per interface:
R1(config)# interface g0/1
R1(config-if)# ip ospf 1 area 0
R1(config-if)# ip ospf network point-to-point            ! no DR election on /30 links
```

Verify	Shows
`show ip ospf neighbor`	Neighbors and state (want FULL)
`show ip ospf interface brief`	Interfaces, area, cost, DR/BDR
`show ip route ospf`	Routes learned by OSPF
`show ip protocols`	Router ID, networks, passive interfaces

Neighbors don't form? Check: same subnet and mask, same area, matching hello/dead timers, unique router IDs, matching MTU, interface not passive, authentication matches, ACL not blocking OSPF.

Wildcard mask = 255.255.255.255 − subnet mask (e.g. /24 → 0.0.0.255, /30 → 0.0.0.3).

12.7 EIGRP and BGP (awareness)

- **EIGRP:** Cisco advanced distance vector, AD 90, metric based on bandwidth + delay, fast convergence (feasible successors).
- **BGP:** the routing protocol of the internet. Connects **autonomous systems (AS)**. Chooses paths by policy and attributes (AS path, local preference...). Used by ISPs, large companies, and cloud hybrid links (AWS Direct Connect, VPN).

12.8 First Hop Redundancy (FHRP)

Problem: hosts have **one** default gateway. If that router fails, they lose connectivity. Solution: two routers share a **virtual IP** (the gateway) and virtual MAC; one is active, the other takes over.

Protocol	Standard	Roles
HSRP	Cisco	Active / Standby. Default priority 100; higher wins; `preempt` to take back
VRRP	Open	Master / Backup
GLBP	Cisco	Load-balances across several gateways

```
R1(config)# interface g0/0
R1(config-if)# standby 1 ip 192.168.10.1      ! virtual gateway IP for hosts
R1(config-if)# standby 1 priority 110
R1(config-if)# standby 1 preempt
R1# show standby brief
```

13. Transport & Application Protocols

13.1 TCP vs UDP

	TCP	UDP
Connection	3-way handshake: SYN → SYN-ACK → ACK	None
Reliability	Sequence numbers, acknowledgments, retransmission	None
Flow control	Sliding **window** (receiver says how much it can take)	None
Order	Guaranteed	Not guaranteed
Close	FIN / ACK (4-way), or RST to abort	—
Header	20+ bytes	8 bytes
Use	Web, SSH, email, file transfer, databases	DNS, DHCP, VoIP, video, SNMP, NTP, syslog

Port ranges: **0–1023** well-known, **1024–49151** registered, **49152–65535** dynamic/ephemeral (client side).

13.2 Common ports and protocols

Protocol	Port	Transport	Purpose
FTP	20/21	TCP	File transfer (plain text)
SSH / SFTP / SCP	22	TCP	Secure remote login / file transfer
Telnet	23	TCP	Remote login, plain text — avoid
SMTP	25 (587 submission)	TCP	Sending email
DNS	53	UDP (TCP for large/zone transfers)	Name resolution
DHCP	67 server / 68 client	UDP	Address assignment
TFTP	69	UDP	Simple file transfer (device images, configs)
HTTP / HTTPS	80 / 443	TCP (HTTP/3 uses UDP/QUIC)	Web
POP3 / IMAP	110 / 143 (995 / 993 secure)	TCP	Reading email
NTP	123	UDP	Time sync
SNMP	161 (traps 162)	UDP	Device monitoring
LDAP / LDAPS	389 / 636	TCP	Directory
Syslog	514	UDP	Log messages

Protocol	Port	Transport	Purpose
RDP	3389	TCP	Windows remote desktop
BGP	179	TCP	Internet routing
RADIUS / TACACS+	1812-1813 / 49	UDP / TCP	Network login (AAA)

14. IP Services

14.1 DHCP

DORA: Discover (broadcast) → Offer → Request → Acknowledge. Gives IP, mask, default gateway, DNS, lease time.

```
R1(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.20    ! keep for static devices
R1(config)# ip dhcp pool SALES
R1(dhcp-config)# network 192.168.10.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.10.1
R1(dhcp-config)# dns-server 8.8.8.8
R1(dhcp-config)# lease 7
R1# show ip dhcp binding
```

DHCP relay: DHCP Discover is a broadcast and routers don't forward broadcasts. If the server is in another subnet, configure a relay on the gateway interface:

```
R1(config)# interface g0/0.10
R1(config-subif)# ip helper-address 10.0.0.50
```

Client side: `ipconfig /release`, `ipconfig /renew` (Windows), `dhclient` (Linux).

14.2 DNS

Resolves names → IPs. Recursive resolver asks Root → TLD → Authoritative. Records: A, AAAA, CNAME, MX, TXT, NS, PTR (reverse lookup), SOA. Cache time is controlled by **TTL**. On a Cisco device: `ip name-server 8.8.8.8` and `ip domain-lookup`.

14.3 NAT (Network Address Translation)

Term	Meaning
Inside local	Private IP of the inside host (192.168.10.5)
Inside global	Public IP representing it on the internet (203.0.113.10)
Outside global	The real public IP of the remote server
Static NAT	One private ↔ one public (servers that must be reachable)
Dynamic NAT	Private IPs use addresses from a public pool
PAT (NAT overload)	Many private IPs share **one** public IP, distinguished by port numbers. Used by almost every home and office

```
R1(config)# interface g0/0
R1(config-if)# ip nat inside
R1(config)# interface g0/1
R1(config-if)# ip nat outside
R1(config)# access-list 1 permit 192.168.0.0 0.0.255.255
R1(config)# ip nat inside source list 1 interface g0/1 overload       ! PAT
R1(config)# ip nat inside source static 192.168.10.5 203.0.113.10     ! static NAT for a server
R1# show ip nat translations
R1# show ip nat statistics
```

14.4 NTP

Keeps device clocks in sync. Correct time is essential for logs, certificates, authentication and troubleshooting. **Stratum** = distance from the reference clock (lower is better).

```
R1(config)# ntp server 192.0.2.10
R1(config)# clock timezone IST 5 30
R1# show ntp status
R1# show ntp associations
```

14.5 SNMP

Monitoring system (**NMS**) polls devices (**agents**) for data (CPU, interface traffic) stored in the **MIB** (identified by **OIDs**). Devices can also send **traps/informs**. Use **SNMPv3** (authentication + encryption); v1/v2c use plain-text community strings.

14.6 Syslog

Devices send log messages to a central server (UDP 514). Severity levels — lower number = more severe:

Level	Name	Level	Name
0	Emergency	4	Warning
1	Alert	5	Notification
2	Critical	6	Informational
3	Error	7	Debugging

Memory trick: **E**very **A**wesome **C**isco **E**ngineer **W**ill **N**eed **I**ce-cream **D**aily.

```
R1(config)# logging host 10.0.0.60
R1(config)# logging trap warnings          ! send levels 0–4
R1(config)# service timestamps log datetime msec
```

14.7 QoS (Quality of Service)

Gives priority to important traffic (voice, video) when links are congested.

Step	Meaning
Classification	Identify traffic (by ACL, application, port)
Marking	Label it: **DSCP** in the IP header (EF = 46 for voice), CoS in the 802.1Q tag
Queuing	Separate queues; **LLQ** gives voice a strict-priority queue
Policing	Drops/re-marks traffic above a rate
Shaping	Buffers traffic to smooth it to a rate
Congestion avoidance	Drop some packets early before queues fill (WRED)

Voice needs: one-way delay < 150 ms, jitter < 30 ms, loss < 1%.

14.8 File transfer for devices

TFTP (simple, no authentication), FTP, SCP/SFTP (secure). Backup a config: `copy running-config tftp:` or `copy running-config scp:`.

14.9 CDP and LLDP

Discover directly connected neighbors (device type, port, IP). CDP = Cisco, LLDP = open standard. Disable on untrusted/internet-facing ports (they reveal information): `no cdp enable`, `no lldp transmit`.

15. Wireless (Wi-Fi)

15.1 Standards and bands

Standard	Name	Band	Max speed (theoretical)
802.11n	Wi-Fi 4	2.4 / 5 GHz	600 Mbps
802.11ac	Wi-Fi 5	5 GHz	~3.5 Gbps
802.11ax	Wi-Fi 6 / 6E	2.4 / 5 / (6E: 6) GHz	~9.6 Gbps
802.11be	Wi-Fi 7	2.4 / 5 / 6 GHz	~46 Gbps

Band	Range	Interference	Non-overlapping channels
2.4 GHz	Longer	High (Bluetooth, microwaves)	1, 6, 11
5 GHz	Shorter	Lower	Many
6 GHz	Shortest	Lowest	Many (newest devices only)

15.2 Terms

Term	Meaning
SSID	The network name
BSSID	MAC address of the AP radio
BSS / ESS	One AP's cell / several APs with the same SSID (roaming)
Autonomous AP	Configured individually
Lightweight AP + WLC	APs controlled centrally by a Wireless LAN Controller via **CAPWAP** tunnels
Cloud-managed AP	Managed from a cloud dashboard (e.g. Meraki)
AP modes	Local (serve clients), FlexConnect (switch traffic locally at branches), Monitor, Sniffer, Bridge
RSSI / SNR	Signal strength / signal-to-noise ratio. Higher SNR = better

15.3 Wireless security

Method	Status
Open / WEP	Insecure — never use
WPA (TKIP)	Obsolete
WPA2-Personal (PSK, AES-CCMP)	OK for home/small office
WPA2/WPA3-Enterprise (802.1X + RADIUS)	Each user has own credentials — use in companies
WPA3-Personal (SAE)	Stronger than PSK against password guessing

Also: separate guest SSID/VLAN, strong passphrases, management frame protection.

16. Network Security Fundamentals

16.1 Core concepts

Term	Meaning
CIA triad	Confidentiality, Integrity, Availability
Vulnerability / threat / exploit	A weakness / something that could abuse it / the actual attack method
Defense in depth	Many layers of protection
Least privilege	Only the access that is needed
AAA	Authentication (who are you), Authorization (what can you do), Accounting (what did you do)
RADIUS vs TACACS+	RADIUS: open, UDP, encrypts only the password, common for network access. TACACS+: Cisco, TCP 49, encrypts everything, per-command authorization for device admin
802.1X	Port-based access control: device must authenticate (via RADIUS) before the switch port allows traffic

16.2 Common attacks

Attack	Description	Defense
Phishing / social engineering	Tricking users	Training, MFA, email filtering
Malware / ransomware	Malicious software	Endpoint protection, patching, backups
DoS / DDoS	Flooding a service	Rate limiting, DDoS protection services
Man-in-the-middle	Intercepting traffic	Encryption (TLS, VPN), DAI
MAC flooding	Fill the switch MAC table so it floods frames	Port security
DHCP spoofing / starvation	Fake DHCP server / exhaust the pool	DHCP snooping
ARP spoofing / poisoning	Fake ARP replies to redirect traffic	Dynamic ARP Inspection (DAI)
VLAN hopping	Abuse DTP/native VLAN to reach other VLANs	Disable DTP, unused native VLAN, access ports static
Password attacks	Brute force, dictionary	Strong passwords, MFA, lockout, SSH keys

16.3 Access Control Lists (ACLs)

- Ordered list of permit/deny rules, checked **top to bottom, first match wins**.
- Every ACL ends with an invisible `deny any`.
- Applied to an interface **inbound or outbound** (one ACL per interface, per direction, per protocol).

Type	Numbers	Matches	Placement
Standard	1–99, 1300–1999	Source IP only	Close to the **destination**
Extended	100–199, 2000–2699	Source, destination, protocol, ports	Close to the **source**
Named	Any name	Standard or extended; easy to edit	—

```
! Allow the 192.168.10.0/24 network to reach web server 10.0.0.80 on HTTPS only; block other traffic
to it
R1(config)# ip access-list extended WEB-ONLY
R1(config-ext-nacl)# 10 permit tcp 192.168.10.0 0.0.0.255 host 10.0.0.80 eq 443
R1(config-ext-nacl)# 20 deny ip any host 10.0.0.80 log
R1(config-ext-nacl)# 30 permit ip any any
R1(config)# interface g0/0
R1(config-if)# ip access-group WEB-ONLY in
R1# show access-lists

! Restrict SSH management access to the admin subnet
R1(config)# ip access-list standard MGMT
R1(config-std-nacl)# permit 10.10.10.0 0.0.0.255
R1(config)# line vty 0 15
R1(config-line)# access-class MGMT in
```

16.4 Layer 2 security

```
! Port security: allow max 2 MACs, learn them, shut the port on violation
SW1(config)# interface f0/5
SW1(config-if)# switchport mode access
SW1(config-if)# switchport port-security
SW1(config-if)# switchport port-security maximum 2
SW1(config-if)# switchport port-security mac-address sticky
SW1(config-if)# switchport port-security violation shutdown     ! or restrict / protect
SW1# show port-security interface f0/5

! DHCP snooping: only trusted ports may send DHCP offers
SW1(config)# ip dhcp snooping
SW1(config)# ip dhcp snooping vlan 10,20
SW1(config)# interface g0/1                                       ! uplink toward real DHCP server
SW1(config-if)# ip dhcp snooping trust

! Dynamic ARP Inspection: validate ARP using the DHCP snooping table
SW1(config)# ip arp inspection vlan 10,20
SW1(config)# interface g0/1
SW1(config-if)# ip arp inspection trust
```

Recover an err-disabled port: fix the cause → `shutdown` → `no shutdown` (or configure `errdisable recovery`). **Good practice:** shut unused ports and put them in an unused VLAN.

16.5 Firewalls, IPS and VPNs

Item	Meaning
Stateful firewall	Tracks connections; allows return traffic automatically
NGFW	Adds application awareness, user identity, IPS, URL filtering, TLS inspection
Zones	Inside (trusted), outside (internet), DMZ (public-facing servers)

Item	Meaning
Site-to-site VPN	Connects two networks over the internet with **IPsec** (encryption + integrity + authentication)
Remote-access VPN	Users connect to the company network (TLS/SSL VPN or IPsec client, e.g. AnyConnect)
GRE	Tunnel without encryption (often combined with IPsec)
Zero trust / ZTNA	Verify every user and device for every access, regardless of network location

IPsec basics: IKE negotiates keys (phase 1 secures the channel, phase 2 creates the data tunnels); ESP encrypts the data.

16.6 Device hardening checklist

`enable secret`, local users with `secret`, SSH v2 only, VTY ACL, AAA with TACACS+/RADIUS, `service password-encryption`, login banner, disable unused services and ports, NTP + syslog, SNMPv3, regular config backups, keep IOS updated.

17. WAN & Internet Connectivity

Technology	Description
Leased line	Dedicated private circuit between two sites. Reliable, expensive
MPLS VPN	Provider's private network connecting company sites; labels forward traffic; supports QoS
Metro Ethernet	Ethernet service across a city from a provider
Broadband internet	Fibre (FTTH), cable, DSL, 4G/5G, satellite. Cheap, "best effort"
Internet VPN	IPsec tunnels over normal internet links
SD-WAN	Software-defined WAN: a central controller manages many links (MPLS + internet + 5G), picks the best path per application, encrypts traffic, simplifies branch setup

ISP connection options: single-homed (1 link, 1 ISP), dual-homed (2 links, 1 ISP), multihomed (2 ISPs), dual-multihomed (2 links to each of 2 ISPs). **Internet edge:** router/firewall + PAT + default route to the ISP; BGP when multihomed with your own public addresses.

18. Data Center, Virtualization & SDN

18.1 Virtualization

Term	Meaning
Hypervisor	Runs virtual machines. Type 1 = on bare metal (VMware ESXi, KVM, Hyper-V). Type 2 = on a host OS (VirtualBox)
Virtual switch	Software switch connecting VMs (vSwitch, Linux bridge, Open vSwitch)
Containers	Share the host kernel; networked through bridges, overlays (Docker, Kubernetes CNI)
VRF	Virtual Routing and Forwarding: several separate routing tables on one router (multi-tenant, isolating networks)
VXLAN	Overlay that carries Layer 2 networks over a Layer 3 underlay (24-bit ID = 16 million segments). Used in data centers and clouds

18.2 Software-Defined Networking (SDN)



Term	Meaning
Control plane	Decides where traffic goes (routing protocols, STP)
Data plane	Actually forwards traffic
Management plane	How you manage the device (SSH, SNMP, APIs)
Traditional networking	Control plane on every device, configured box by box
Controller-based networking	Central controller holds policy and pushes config to many devices
Underlay / overlay / fabric	Physical IP network / virtual networks on top / the whole system
Intent-based networking	You describe business intent ("guests can only reach the internet"); the controller configures devices

18.3 Network automation

Item	Meaning
Why automate	Fewer manual errors, consistency, speed, audit trail in Git
REST API	HTTP methods on resources: GET (read), POST (create), PUT/PATCH (update), DELETE

Item	Meaning
CRUD	Create, Read, Update, Delete ↔ POST, GET, PUT/PATCH, DELETE
JSON	Data format for APIs: `{"interface": "Gi0/1", "enabled": true, "vlans": [10, 20]}`
YAML / XML	Other data formats (Ansible uses YAML; NETCONF uses XML)
YANG	Data **model** describing device configuration and state
NETCONF / RESTCONF	Protocols to read/change config using YANG models (NETCONF over SSH 830; RESTCONF over HTTPS)
Ansible	Agentless automation over SSH/APIs, playbooks in YAML. Push model
Puppet / Chef	Agent-based configuration management. Pull model
Terraform	Infrastructure as code for cloud and some network platforms
Python libraries	Netmiko (SSH to devices), NAPALM (multi-vendor), `requests` (APIs), ncclient (NETCONF)

```
# Example: show interfaces on a Cisco device with Netmiko
from netmiko import ConnectHandler
import os

device = {
    "device_type": "cisco_ios",
    "host": "10.0.0.1",
    "username": os.environ["NET_USER"],       # credentials from environment, never in code
    "password": os.environ["NET_PASS"],
}
with ConnectHandler(**device) as conn:
    print(conn.send_command("show ip interface brief"))
```

```
# Example: Ansible playbook to create VLANs on Cisco switches
- name: Configure VLANs
  hosts: access_switches
  gather_facts: false
  tasks:
    - name: Ensure VLANs exist
      cisco.ios.ios_vlans:
        config:
          - { vlan_id: 10, name: SALES }
          - { vlan_id: 20, name: HR }
        state: merged
```

19. Cloud Networking (AWS mapping)

Traditional network	AWS equivalent
Company network / data center	VPC
Subnet / VLAN	Subnet (one per AZ)
Router + routing table	VPC router + route tables
Internet edge router	Internet Gateway
NAT/PAT router	NAT Gateway
Stateful firewall on a host	Security Group
Stateless ACL on a subnet	Network ACL
Load balancer	ALB (L7) / NLB (L4)
DNS server	Route 53 (+ Resolver for hybrid DNS)
Site-to-site VPN	AWS Site-to-Site VPN (IPsec, BGP or static)
Leased line / MPLS to cloud	AWS Direct Connect
Core / hub router connecting many networks	Transit Gateway
Point-to-point network link	VPC peering
Packet capture / flow logs	VPC Traffic Mirroring / VPC Flow Logs

Cloud design rules: plan non-overlapping CIDRs with on-premises and other VPCs; public subnets only for load balancers/NAT; apps and databases in private subnets; use multiple AZs; Security Groups referencing each other.

Kubernetes networking (for DevOps): every pod gets an IP (CNI plugin: AWS VPC CNI, Calico, Cilium); Services give stable virtual IPs; Ingress handles HTTP routing; NetworkPolicies are the firewall between pods; CoreDNS provides service names.

20. Troubleshooting

20.1 Methods

Method	How
Bottom-up	Start at Layer 1 (cable, link lights), move up. Good for physical/new installs
Top-down	Start at the application, move down. Good when only one app fails
Divide and conquer	Start in the middle (e.g. `ping` = Layer 3) and go up or down based on the result. Fastest in practice
Follow the path	Trace the packet hop by hop: host → switch → gateway → routers → server
Compare	Compare with a working device/config

Process: define the problem → gather information → what changed? → hypothesis → test → fix → verify → document.

20.2 Tools

Tool	Windows	Linux / macOS	Cisco IOS
IP config	`ipconfig /all`	`ip a`	`show ip interface brief`
Reachability	`ping`	`ping`	`ping` (extended: `ping` then options)
Path	`tracert`	`traceroute`, `mtr`	`traceroute`
ARP / neighbors	`arp -a`	`ip neigh`	`show ip arp`, `show mac address-table`
Routes	`route print`	`ip route`	`show ip route`
DNS	`nslookup`	`dig`, `nslookup`	`ping <name>` with `ip name-server`
Ports	`netstat -ano`, `Test-NetConnection -Port`	`ss -tulpn`, `nc -zv`	`telnet <ip> <port>` (as a port test)
Packets	Wireshark	`tcpdump`, Wireshark	Embedded Packet Capture, SPAN port mirroring

Ping results: `Reply` = OK · `Request timed out` = no answer (down, firewall, routing on the return path) · `Destination host unreachable` = a router/host says it has no route or no ARP reply · `TTL expired in transit` = routing loop.

20.3 Common scenarios

Problem	Likely causes and checks
PC has 169.254.x.x	DHCP not reachable: cable/VLAN, DHCP server down, missing `ip helper-address`, pool exhausted, DHCP snooping blocking

Problem	Likely causes and checks
Can ping by IP, not by name	DNS: wrong DNS server, DNS down, record missing (`nslookup`)
Same VLAN works, other VLANs don't	Default gateway wrong/missing on host, inter-VLAN routing (SVI down, subinterface/encapsulation wrong), trunk not allowing VLAN
Two switches, VLAN traffic doesn't pass	Trunk down, VLAN not created on both, VLAN not allowed on trunk, native VLAN mismatch
Port is err-disabled	Port security violation or BPDU Guard → fix cause, `shutdown` / `no shutdown`
Network slow / broadcast storm	Layer 2 loop (STP disabled/misconfigured), duplex mismatch, congested link, errors on interface
OSPF neighbors not forming	Subnet/mask, area, timers, MTU, passive interface, duplicate router ID, authentication
Can reach internet by IP from router but not from LAN	NAT: inside/outside wrong, ACL doesn't match LAN, no default route
Remote site unreachable	Route missing on one side (return path!), VPN down, ACL/firewall
Intermittent drops	Interface errors (CRC), Wi-Fi interference, flapping link, overloaded device

21. Interview Questions (Quick Answers)

Question	Short answer
Difference between a switch and a router?	Switch forwards frames inside a network by MAC (L2); router forwards packets between networks by IP (L3)
What happens when you type a URL in a browser?	DNS lookup → TCP handshake → TLS handshake → HTTP request → server response → page rendered
TCP vs UDP?	TCP: connection, reliable, ordered. UDP: connectionless, fast, no guarantees
What is ARP?	Finds the MAC address for an IP on the local network
What is a VLAN and why use it?	Logical network on a switch; separates broadcast domains for security and organization
What is a trunk?	A link carrying multiple VLANs with 802.1Q tags
Why STP?	Prevents Layer 2 loops when there are redundant links
What is a default gateway?	The router a host uses to reach other networks
Static vs dynamic routing?	Static: manual, simple, no adaptation. Dynamic: routers exchange routes and adapt to failures
OSPF vs BGP?	OSPF: link-state IGP inside one organization. BGP: path-vector EGP between organizations / the internet
What is administrative distance?	Trust value used to choose between routes from different sources; lower wins
NAT vs PAT?	NAT maps IPs; PAT maps many private IPs to one public IP using ports
Public vs private IP?	Public: routable on the internet. Private (RFC 1918): internal only, needs NAT
What is DHCP?	Automatically assigns IP configuration (DORA)
How do you troubleshoot "no internet"?	IP config → gateway ping → ping 8.8.8.8 → DNS → proxy/firewall → application
Standard vs extended ACL?	Standard: source only, near destination. Extended: source/dest/protocol/port, near source
What is a subnet mask?	Shows which part of an IP is network and which is host
How many hosts in a /26?	64 addresses, 62 usable
What is HSRP?	Cisco FHRP: two routers share a virtual gateway IP for redundancy
Security Group vs NACL (AWS)?	SG: stateful, instance level, allow only. NACL: stateless, subnet level, allow and deny
What is SD-WAN?	Centrally managed WAN that uses several link types and chooses paths per application
What is a VPN?	An encrypted tunnel over an untrusted network (IPsec, TLS)